

Network Science

Lecture 05: Social Context and Link Formation

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This lecture extends the network model with context, attributes, and longitudinal processes.

Social Context and Link Formation

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Focus

- homophily
- selection and socialization
- affiliation networks and closure beyond person-person ties
- online link formation and spatial segregation

Module 1

Homophily

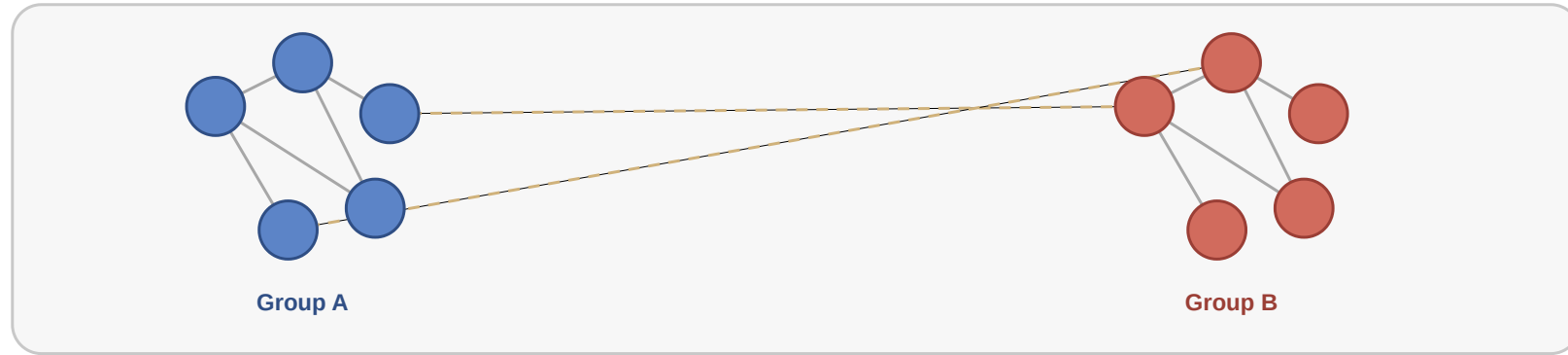
Guiding Question: Do people become friends because they are similar, or do they become similar because they are friends?

Formal Intuition

Homophily means that edge formation is statistically biased toward similarity in node attributes.

Visual Intuition

Homophily as similarity-biased tie formation



Solid edges are frequent within-group ties; dashed edges are rarer cross-group ties.

Interpretation

Friendship ties are often biased toward similarity in attributes such as age, background, beliefs, or activity.

Why This Matters

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- it shapes who meets whom

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- it changes which information spreads where

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- it shapes who meets whom
- it changes which information spreads where
- it complicates causal interpretation of network data

Mini-Exercise 5.A: Detecting Homophily

In a school of 200 students, 60% are in track A and 40% in track B. Of all friendships, 78% are within-track.

1. What fraction of friendships would you expect to be within-track if friendships were random with respect to track?
2. Is the observed 78% evidence of homophily? Why or why not?

Mini-Exercise 5.A: Solution

Yes. Observed 78% is well above the 52% random baseline, indicating that friendships are biased toward within-track. The lesson: declaring homophily requires a counterfactual — "more than expected if links were random with respect to the attribute".

Mini-Exercise 5.A: Solution

1. Random baseline:

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Mini-Exercise 5.A: Solution

1. Random baseline:

$$P(\text{same track}) = 0.6^2 + 0.4^2 = 0.36 + 0.16 = 0.52$$

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Mini-Exercise 5.A: Solution

1. Random baseline:

$$P(\text{same track}) = 0.6^2 + 0.4^2 = 0.36 + 0.16 = 0.52$$

2. Evidence of homophily: Yes. Observed 78% is well above the 52% random baseline, indicating that friendships are biased toward within-track. The lesson: declaring homophily requires a counterfactual — "more than expected if links were random with respect to the attribute".

Takeaway

Homophily is not only a descriptive pattern. It is a mechanism that shapes network formation and measurement.

Module 2

Selection and Socialization

Guiding Question: How can we distinguish selection from social influence?

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$\text{homophily} = \text{selection} + \text{socialization} + \text{context}$

Core Distinction

- : people choose similar others
- : people become more similar after linking
- : both may reflect a third factor (shared context)

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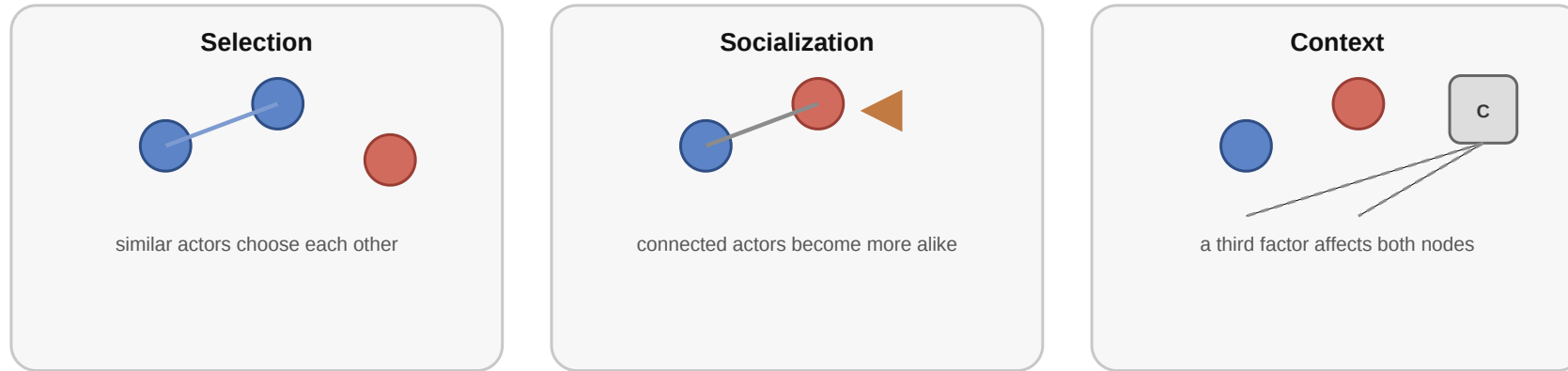
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Core Distinction

- **selection:** people choose similar others
- **socialization:** people become more similar after linking
- **correlation:** both may reflect a third factor (shared context)

Example

Why observed similarity can arise in different ways



Mini-Exercise 5.B: Designing a Test

You observe that smokers tend to befriend other smokers. You want to determine whether this is selection (smokers prefer smoker friends) or socialization (people start smoking because their friends do).

1. What kind of data would you need beyond a single snapshot of friendships?
2. Sketch one observation that would favor each explanation.

Mini-Exercise 5.B: Solution

Longitudinal observations — friendships and smoking status at multiple time points, ideally with newly formed ties and newly adopted behavior visible.

- : New friendships disproportionately form between people who already share smoking status.
- : People who befriend smokers become more likely to start smoking themselves over the next observation period.

Mini-Exercise 5.B: Solution

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2. Distinguishing observations:

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Mini-Exercise 5.B: Solution

1. Required data: Longitudinal observations — friendships and smoking status at multiple time points, ideally with newly formed ties and newly adopted behavior visible.

2. Distinguishing observations:

- *Selection:* New friendships disproportionately form between people who already share smoking status.
- *Diffusion:* People who befriend smokers become more likely to start smoking themselves over the next observation period.

Mini-Exercise 5.B: Solution

1. Required data: Longitudinal observations — friendships and smoking status at multiple time points, ideally with newly formed ties and newly adopted behavior visible.

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- *Socialization:* People who befriend smokers become more likely to start smoking themselves over the next observation period.

Takeaway

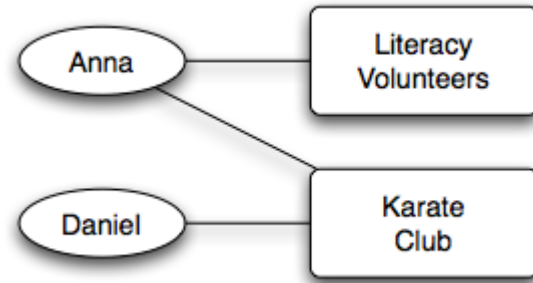
Cross-sectional graphs rarely settle the causal question. Longitudinal network data is much more informative.

Module 3

Affiliation Networks

Guiding Question: How do we represent the social settings in which ties form?

Bipartite Affiliation



Source: Easley & Kleinberg 2010

Affiliation networks distinguish at least two node types:

- persons
- foci such as groups, locations, classes, or organizations

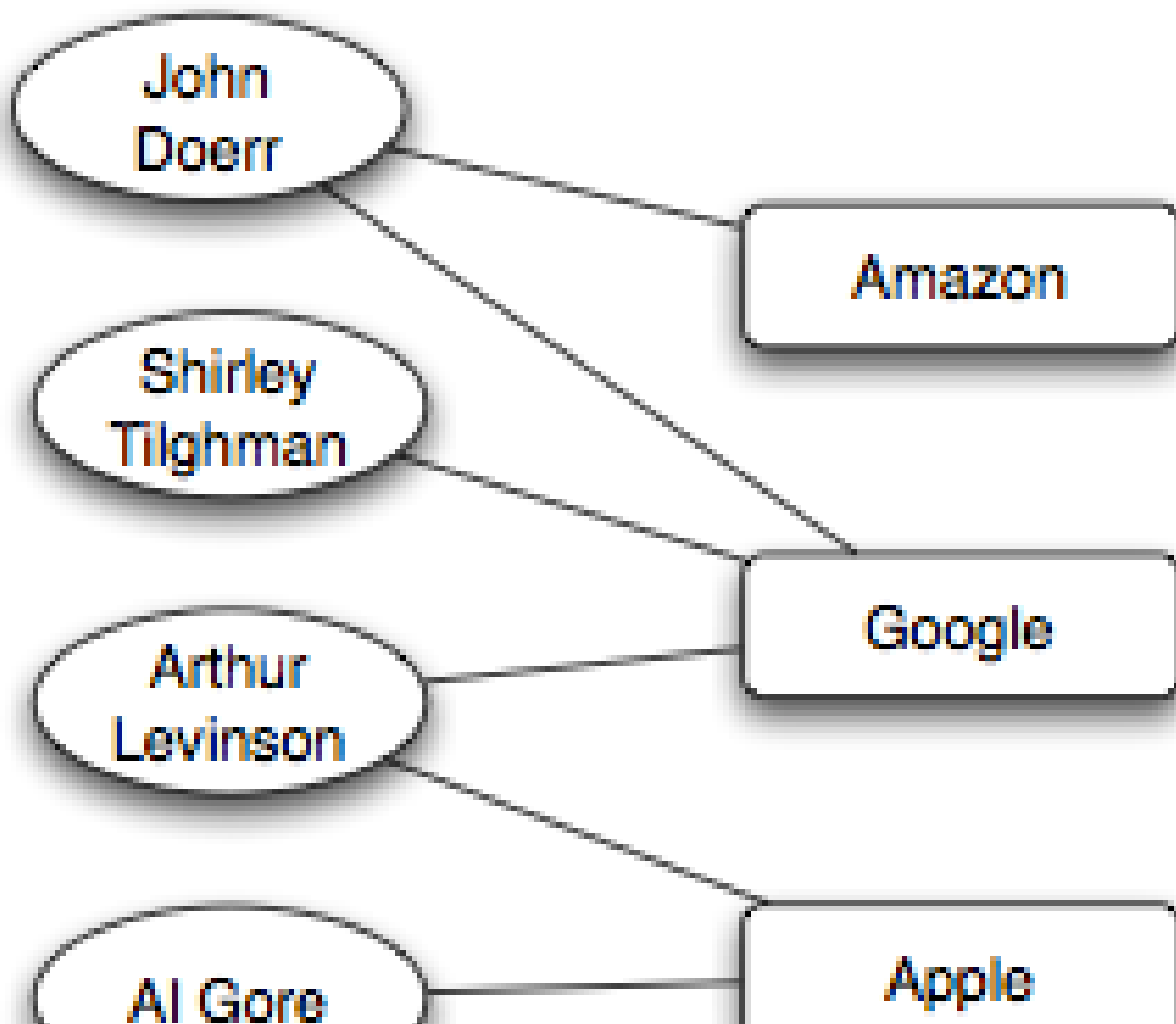
Formal Definition

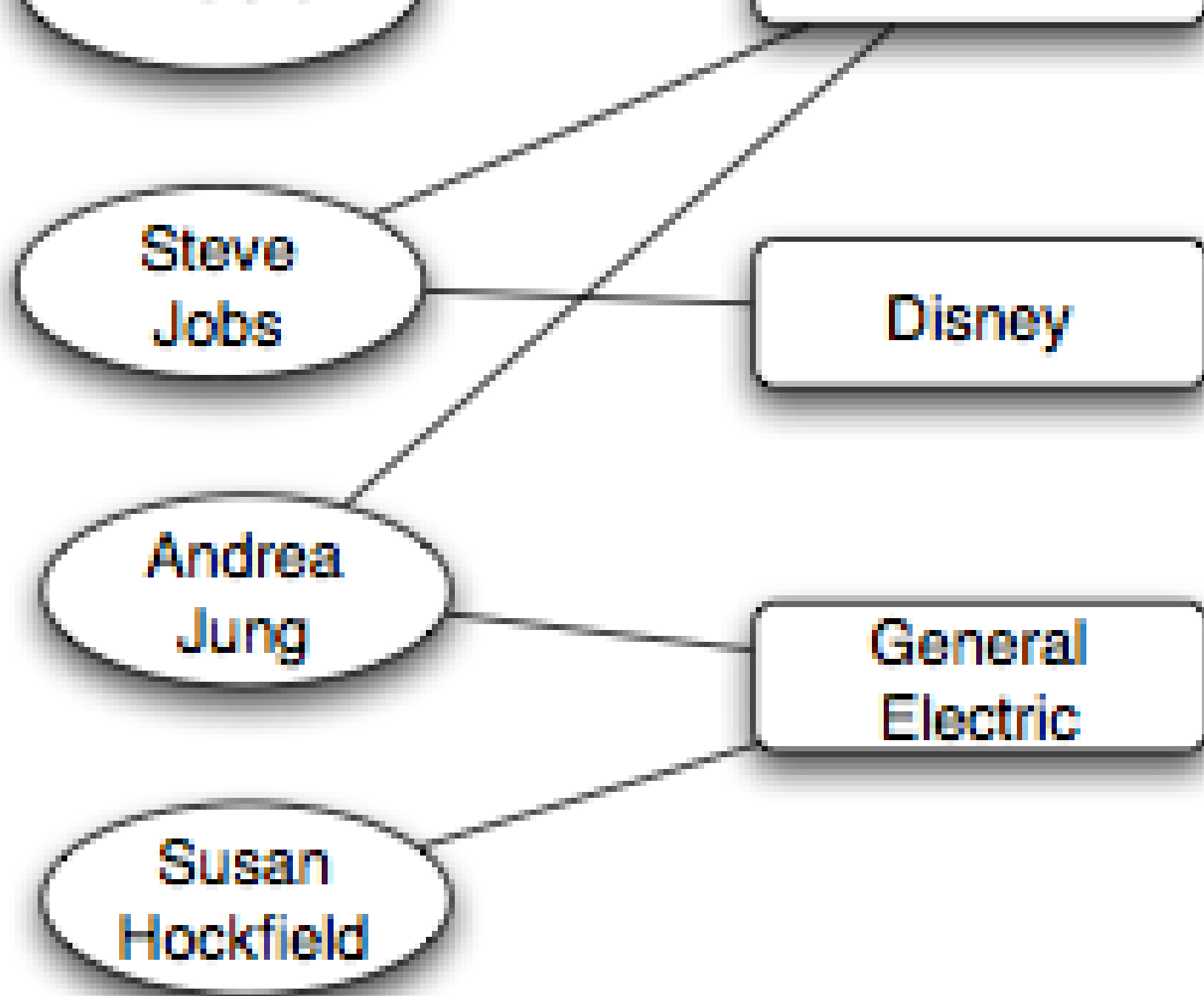
An affiliation network is naturally modeled as a bipartite graph:

- one node set for actors
- one node set for foci
- edges only between actors and foci

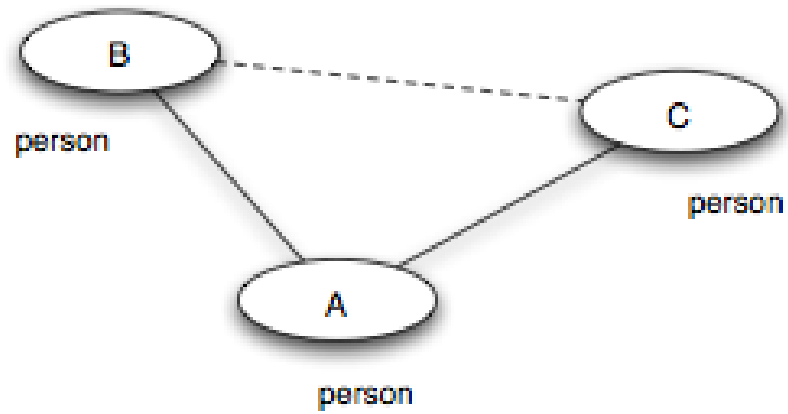
Example



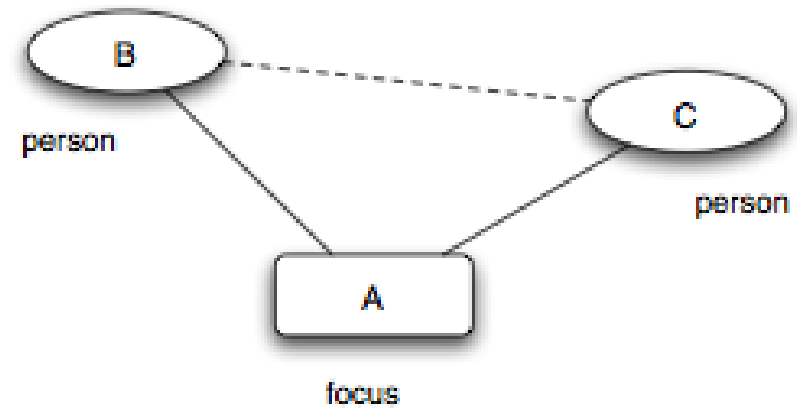




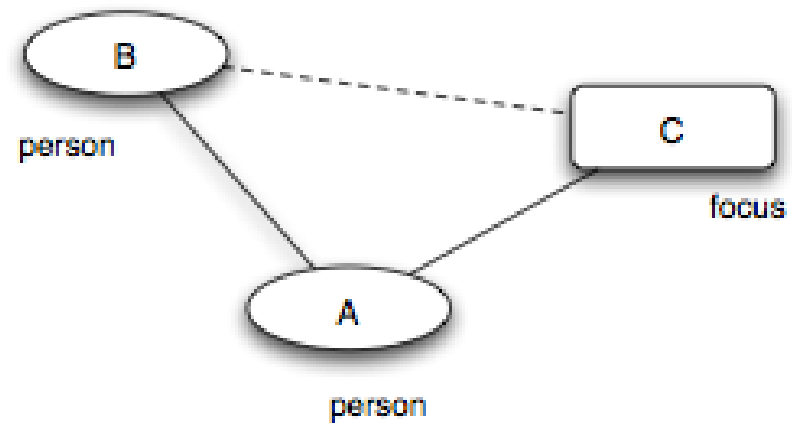
Closure Processes



(a) *Triadic closure*



(b) *Focal closure*



Source: Easley & Kleinberg 2010

Mini-Exercise 5.C: Identify the Closure Type

For each scenario, identify whether triadic, focal, or membership closure is at work.

1. Two students take the same elective and become friends.
2. A new graduate student joins the same lab as her advisor's other student, who introduces her.
3. A friend invites you to their book club, and you join.

Mini-Exercise 5.C: Solution

1. — the elective is the shared focus that brings them into contact.
2. — the advisor is the shared friend that closes the triangle.
3. — your friend brings you into a focus you did not previously belong to.

Mini-Exercise 5.C: Solution

1. **Focal closure** — the elective is the shared focus that brings them into contact.
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Mini-Exercise 5.C: Solution

1. **Focal closure** — the elective is the shared focus that brings them into contact.
2. **Triadic closure** — the advisor is the shared friend that closes the triangle.
3. **Membership closure** — your friend brings you into a focus you did not previously belong to.

Takeaway

Affiliation networks make context explicit and let us formalize focal closure and membership closure alongside ordinary triadic closure.

Module 4

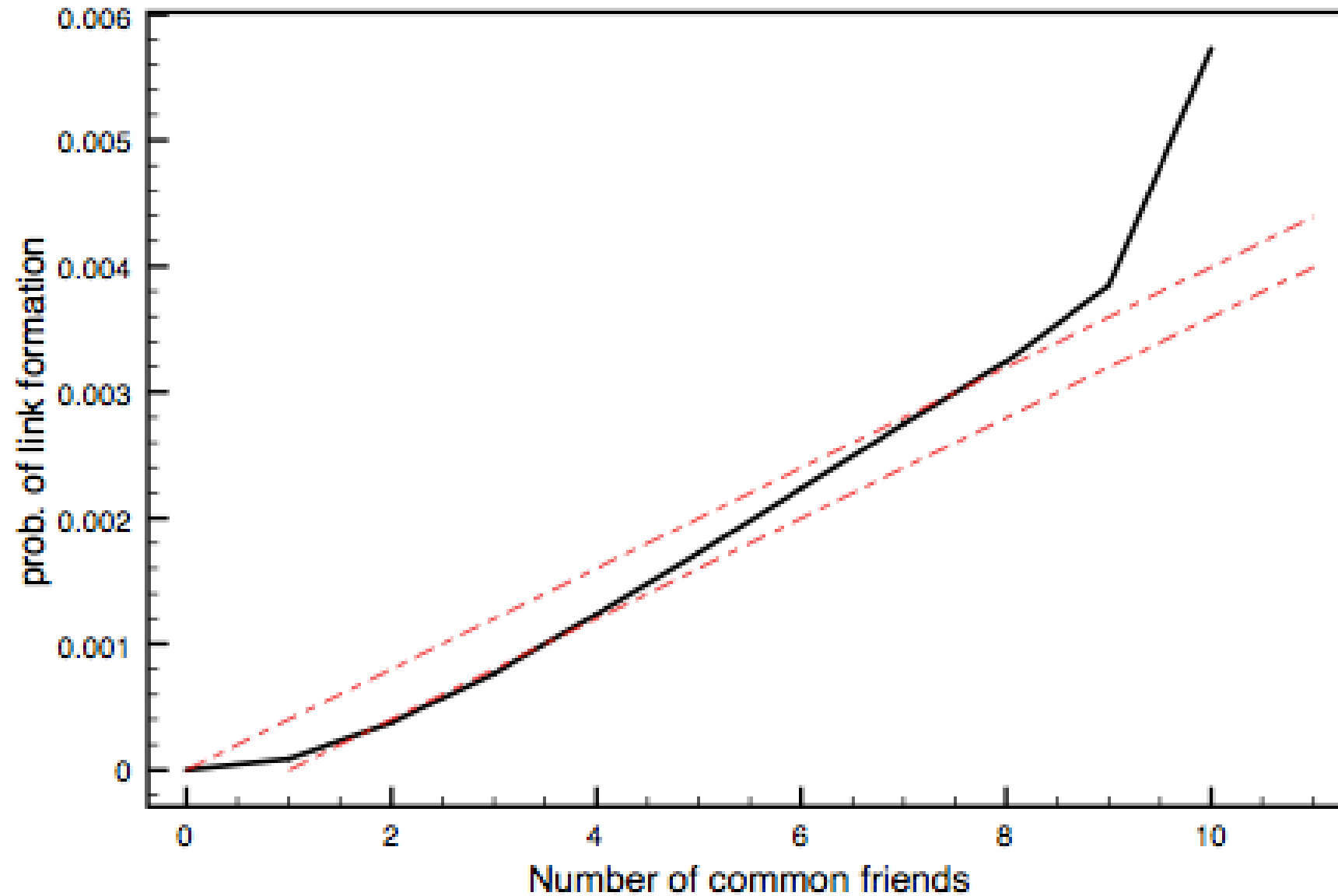
Online Link Formation

Guiding Question: What can large online datasets teach us about how links form over time?

Formal Intuition

Link formation can be studied longitudinally by observing when new edges appear and which local conditions precede them.

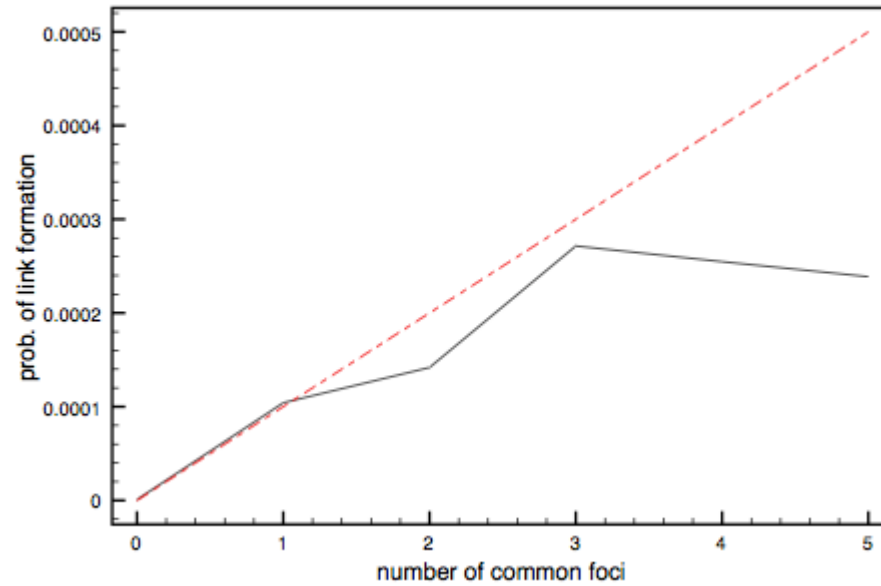
Triadic Closure in Email



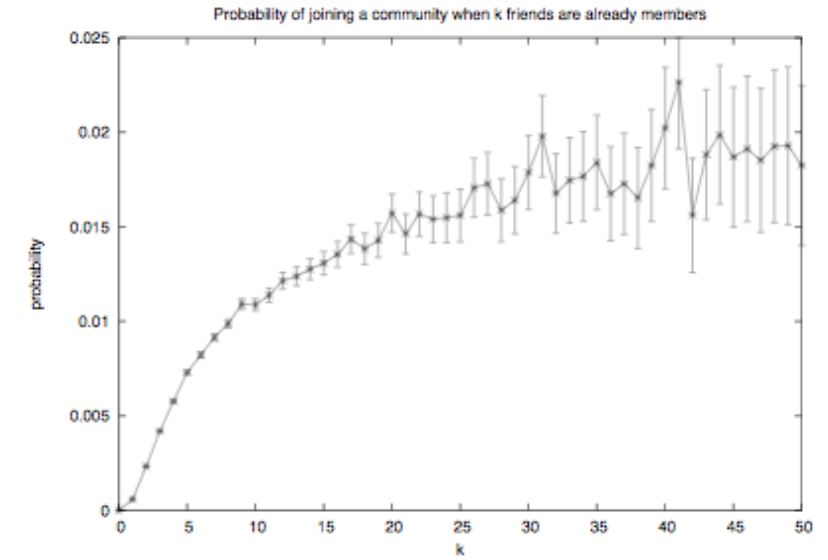
Source: Easley & Kleinberg 2010

The probability of a new edge rises with the number of shared contacts.

Focal and Membership Closure



Source: Easley & Kleinberg 2010



Source: Easley & Kleinberg 2010

Interpretation

Shared foci and shared friends both matter, but often with diminishing returns rather than simple linear growth.

Mini-Exercise 5.D: Reading a Closure Curve

A study finds that the probability of two coworkers becoming Facebook friends rises from 1% (no shared friends) to 20% (one shared friend) to 30% (two shared friends) to 33% (three or more).

1. Describe the shape of the curve.
2. What does the diminishing-returns pattern suggest about the mechanism of closure?

Mini-Exercise 5.D: Solution

Steep initial rise from 1% to 20% with the first shared friend, then much smaller gains (30%, 33%) for additional shared friends.

Diminishing returns suggest that the first shared contact captures most of the available "social credit" — the new opportunity, the trust, the introduction. Additional shared friends reinforce but do not multiply this effect. The mechanism is not purely additive.

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Mini-Exercise 5.D: Solution

- 1. Shape:** Steep initial rise from 1% to 20% with the first shared friend, then much smaller gains (30%, 33%) for additional shared friends.
- 2. Mechanism interpretation:** Diminishing returns suggest that the first shared contact captures most of the available "social credit" — the new opportunity, the trust, the introduction. Additional shared friends reinforce but do not multiply this effect. The mechanism is not purely additive.

Takeaway

Online traces make network formation measurable, but the mechanism still needs interpretation.

Module 5

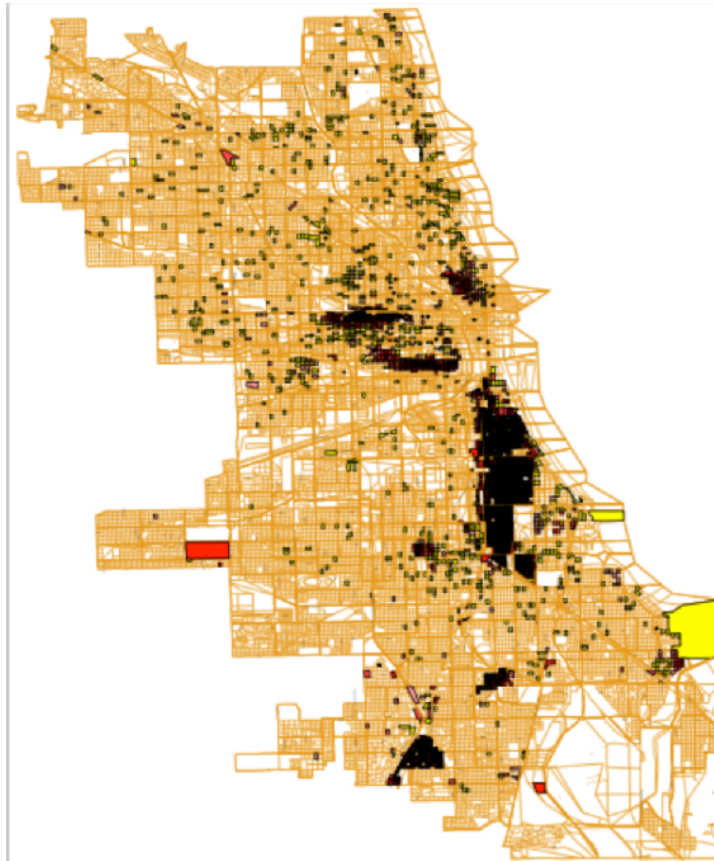
Spatial Segregation

Guiding Question: Can weak local preferences generate strong global segregation?

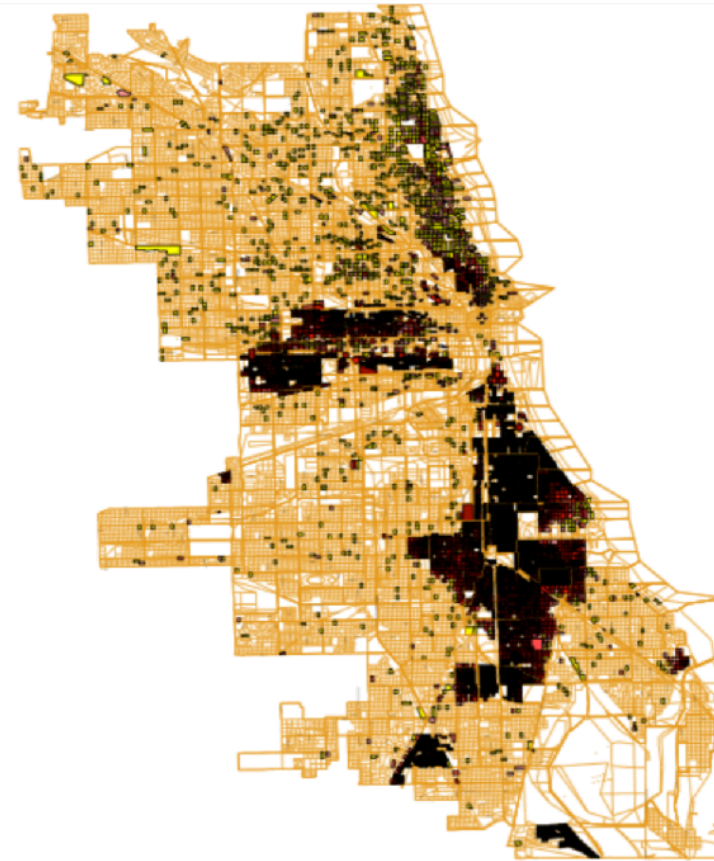
Formal Intuition

Segregation can emerge from repeated local updates even when each individual preference is mild.

Empirical Motivation



(a) *Chicago, 1940*



(b) *Chicago, 1960*

Source: Easley & Kleinberg 2010

Schelling Interpretation

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- agents prefer some minimum level of similar neighbors

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- agents prefer some minimum level of similar neighbors
- dissatisfied agents move locally
- many local moves can produce sharply segregated global patterns

Why This Matters

The lesson is not limited to geography. Many network processes transform mild local rules into strong large-scale structure.

Mini-Exercise 5.E: Schelling in Networks

Translate Schelling's spatial model into a network setting.

1. What plays the role of "neighbor" in a network rather than a grid?
2. What plays the role of "moving"?
3. Predict: would you expect online recommendation algorithms to amplify or weaken homophily? Why?

Mini-Exercise 5.E: Solution

A graph-adjacent node — your immediate contacts.

Rewiring — dropping ties to dissimilar contacts and forming new ones with similar contacts.

3. Algorithmic amplification: Recommendation systems typically exhibit homophily. They surface similar content and similar people, which accelerates the Schelling-style rewiring toward more similar neighborhoods. The local rule ("recommend what is similar") is mild, but the global outcome (filter bubbles) is sharp.

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Takeaway

Global polarization can emerge without anyone explicitly aiming for it. Mild local preferences plus repeated local updates produce sharp large-scale structure.

Wrap-Up

- links often reflect both social similarity and social context
- longitudinal data helps separate formation from influence
- simple local preferences can generate strong global patterns

Looking Ahead: Lecture 06

If signs (positive and negative) and not just connections matter, how do whole social structures stabilize or destabilize? Structural balance theory turns the question of friend-and-enemy relations into a graph-theoretic claim with measurable predictions.

Image Credits & References

Easley, David and Kleinberg, Jon (2010). Networks, Crowds, and Markets: Reasoning About a Highly Connected World. *Cambridge University Press*. <https://www.cs.cornell.edu/home/kleinber/net-works-book/>